# EFRAIN RIVAS

● ENTERPRISE CLOUD & KUBERNETES ARCHITECT

SENIOR CLOUD ARCHITECT & INFRASTRUCTURE SYSTEMS ENGINEER

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## // 01 PROFESSIONAL PROFILE & ARCHITECTURAL FOCUS

Results-driven **Senior Cloud Architect and Systems Engineer** with 20+ years of hands-on enterprise infrastructure experience designing, modernizing, and automating mission-critical systems across Microsoft Azure, Kubernetes, Sovereign AI/LLM platforms, Active Directory Domain Services, Linux, and multi-hypervisor virtualization clusters. Proven architectural track record deploying global multi-site Active Directory forests across 7 global data centers, engineering 5-stage financial CI/CD release pipelines with 99.999% SLA uptime, orchestrating a 6-node high-availability Kubernetes (K3s) cluster (36 vCPUs / 56GB RAM, 50+ pods across 12 namespaces), engineering dual-GPU local AI & RAG inference platforms, and implementing zero-touch endpoint provisioning via Microsoft Intune and Windows Autopilot. Creator of high-performance open-source platforms and engineering tools (**NovaFlix, AegisDNS, Pulsearr, Renamarr, PrintForge, Proxmox Cluster Suite, 58+ Intune remediation modules**) with deep expertise in PowerShell, Python, Proxmox VE clustering, and enterprise identity governance.

## // 02 CORE TECHNICAL COMPETENCIES & SYSTEMS MATRIX

### ▶ CLOUD & DIRECTORY ARCHITECTURE

`Microsoft Azure` `Entra ID (Azure AD)` `Active Directory DS`
`Hybrid Identity (PHS/PTA)` `VNets / NSGs` `Azure Files`
`Group Policy (GPO)` `Exchange Online` `M365 Tenant Admin`

### ▶ CONTAINERS & HA KUBERNETES

`6-Node K3s / K8s HA` `Dual Control-Plane etcd`
`Traefik & NPM Ingress` `Cloudflared DaemonSets`
`50+ Pods / 12 Namespaces` `NFS Persistent Storage`
`Klipper ServiceLB` `Sysctl Kernel Tuning`

### ▶ VIRTUALIZATION & HYPERVISORS

`Proxmox VE (5-Node HA)` `VMware vSphere / ESXi` `Linux KVM`
`Citrix XenServer / XCP-NG` `Microsoft Hyper-V`
`AVX2 Host Passthrough` `Zero-Balloon Memory Tuning`
`LXC to K8s Migration` `SAN / NFS / Ceph`

### ▶ SOVEREIGN AI, DEVOPS & AUTOMATION

`PowerShell 7+` `Python (Asyncio/FastAPI)`
`Dual-GPU AI (Ollama/RAG)` `Open WebUI & Paperclip AI`
`CI/CD Release Pipelines` `CUDA 12.4 / ChromaDB`
`Ansible & Terraform` `Git / Systemd Daemons`

### ▶ ENDPOINT & DEVICE GOVERNANCE

`Microsoft Intune (MDM/MAM)` `Windows Autopilot`
`SCCM / MECM` `Win32 Packaging (.intunewin)`
`Proactive Remediations` `BitLocker Key Escrow`
`Patch Compliance Governance` `Windows 10/11 & macOS`

### ▶ ZERO TRUST, SSO, DNS & OBSERVABILITY

`Authentik Enterprise SSO` `Checkmk & Reconya L2`
`Pi-hole v6 DoH / Unbound` `Nebula-Sync 3-Node DNS`
`Automated SSL (DNS-01)` `Conditional Access / PIM`
`SAML 2.0 / OAuth / FIDO2` `Defender for Cloud`

## // 03 ENTERPRISE INFRASTRUCTURE EXPERIENCE

**Senior IT Infrastructure Consultant** ● **O.S. Consulting, Inc.** Remote | 2025 – Present

- **6-Node HA Kubernetes (K3s) Orchestration**: Architected and deployed a 6-node HA K3s cluster (2 etcd/control-plane nodes + 4 workers, 36 vCPUs, 56GB RAM) hosting 50+ pods across 12 production namespaces (Authentik SSO, Checkmk, Pi-hole DoH/DNSSEC, Portainer, Tomcat 10.1, Reconya, PrintForge, Arr media suite); implemented 4-node Cloudflare Tunnel DaemonSets, Traefik & NPM ingress, kernel sysctl optimization, and shared NFS storage.
- **Sovereign AI & Dual-GPU RAG Infrastructure**: Engineered a dedicated dual-GPU AI platform (NVIDIA RTX 5070 12GB + RTX 3060 12GB) running Ollama local LLMs, Open WebUI with isolated CUDA 12.4 PyTorch acceleration for ChromaDB vector RAG embeddings & Faster-Whisper STT, and Paperclip AI multi-agent control plane orchestration.
- **Proxmox VE Fleet & Workload Modernization**: Administer a 5-node Proxmox VE HA cluster, executing live LXC-to-Kubernetes migrations (Apache Tomcat 10.1, Reconya L2 discovery), configuring 12-core AVX2 host passthrough and 4GB RAM tmpfs transcode buffers, and eliminating memory ballooning contention.
- **Tri-Node DNSSEC/DoH & Automated SSL Lifecycle**: Engineered a 3-node Pi-hole v6, Unbound recursive DNSSEC, and Cloudflare DoH sidecar cluster synced via Nebula-Sync; built automated Let's Encrypt wildcard SSL renewal pipelines via Cloudflare DNS-01 APIs and Kubernetes CronJobs.
- **Hybrid Identity & Modern Endpoint Governance**: Architect hybrid AD to Entra ID synchronizations with PHS, Seamless SSO, and Conditional Access; modernize legacy SCCM/GPO environments to Microsoft Intune with zero-touch Windows Autopilot.
- **Tier 3 Root Cause Analysis (RCA) & Observability**: Deployed Checkmk enterprise monitoring and Reconya continuous Layer-2 ARP/ICMP discovery; serve as principal escalation architect for complex hypervisor, directory, and network incidents.

### Senior Systems Administrator & Infrastructure Architect • BlueMatrix

Durham, NC | 2018 – 2025

- **Global Multi-Site Active Directory Forest**: Architected, built, and maintained enterprise Active Directory forest and domain infrastructure across 7 global data centers and regional offices; tuned multi-site replication topology, site-cost weighting, and distributed DNS to guarantee sub-second authentication worldwide.
- **Multi-Hypervisor Virtualization Clusters**: Administered and tuned high-density virtualization clusters across Linux KVM, Citrix XenServer, VMware vSphere/ESXi, and XCP-NG; orchestrated shared SAN/NFS arrays, vCPU scheduling, dynamic memory allocations, and automated snapshot replication.
- **Azure Cloud Infrastructure & Automation**: Spearheaded phased migrations transitioning mission-critical on-premises systems to Microsoft Azure; architected VNets, Network Security Groups (NSGs), Azure Virtual Machines, and custom Azure Automation Runbooks to govern automated compute scaling and patch compliance.
- **Automated Identity Lifecycle Engineering**: Developed robust PowerShell automation workflows connecting HR systems with AD DS and Entra ID, completely automating provisioning, RBAC group delegation, Microsoft 365 licensing, and zero-day offboarding security revocation.

### DevOps Engineer & Systems Administrator • StreamingEdge (Tradition America)

New York, NY (Hybrid) | 2013 – 2018

- **Financial CI/CD Release Pipeline Engineering**: Directed release engineering across a structured 5-stage deployment pipeline (DEV ➔ QA ➔ UAT ➔ Staging ➔ Production) for institutional financial trading systems, coordinating zero-downtime off-hours application releases and configuration verification.
- **Mission-Critical Systems Reliability (99.999% SLA)**: Engineered and administered high-availability Windows Server (2012–2022) and Linux (RHEL, CentOS, Ubuntu) server farms hosting high-throughput financial transaction engines, consistently maintaining five-nines (99.999%) SLA uptime.
- **Clustered Enterprise Virtualization**: Managed fault-tolerant virtualization clusters across Microsoft Hyper-V and VMware vSphere, configuring multi-NIC teaming, virtual switch isolation, host failover clustering, and disaster recovery replication.
- **Enterprise Networking & Perimeter Security**: Designed and maintained high-throughput LAN/WAN routing, core switching, hardware firewalls, site-to-site IPsec VPN tunnels, and enterprise telecommunications infrastructure.

### Senior Network Administrator • Digital Network Development

Peoria, IL | 1999 – 2011

- **Enterprise Windows Server Operations**: Commissioned, upgraded, and administered physical enterprise servers spanning Windows NT 4.0, 2000, 2003, and 2008, managing Active Directory, Group Policy Objects, DNS, DHCP, and enterprise storage.
- **Telecommunications & Core Networking**: Engineered, programmed, and maintained managed Cisco switches, routers, multi-building wireless links, T1/T3 telecommunications circuits, and enterprise VOIP telephony platforms.
- **Disaster Recovery & Tier 3 Escalation**: Managed enterprise tape and NAS backup routines, storage arrays, OS bare-metal imaging (WDS/MDT), and routine patch compliance across distributed server farms.

## // 04 FLAGSHIP ENGINEERING PROJECTS & OPEN SOURCE TOOLS

**NovaFlix (Creator & Architect)** — github.com/digdast/novaflix
Sovereign open media platform & home theater engine (Plex/Jellyfin alternative) with 4K HDR Direct Play, HTTP 206 Range streaming, built-in remote access, and SyncPlay.

**AegisDNS (Creator & Architect)** — github.com/digdast/aegisdns
Recursive DNS resolver & RFC 8484 DNS-over-HTTPS (DoH) engine traversing root hints with zero telemetry, 227k+ threat sinkhole rules, and real-time cyber UI.

**6-Node HA K3s Cluster & Ingress** — resume.ertechlab.com
6-node K3s cluster (36 vCPUs / 56GB RAM, 50+ pods across 12 namespaces) with Authentik SSO, Checkmk, 4-node Cloudflared DaemonSet, and 4-replica HA web hosting.

**Sovereign Dual-GPU AI & RAG** — Proxmox / AI Lab
Dual-GPU LLM & RAG platform (RTX 5070 + RTX 3060) running Ollama, Open WebUI with CUDA 12.4 ChromaDB vector embeddings, and Paperclip AI agent orchestration.

**Pulsearr & Renamarr (Creator)** — github.com/digdast/pulsearr
Async Python media daemon unifying Sonarr/Radarr/Prowlarr in 85MB RAM (FastAPI/SQLite WAL) + zero-config IMDb/TVMaze media organizer with 1-click undo.

**PrintForge 3D & Intune Catalog** — 3dprinting.ertechlab.com
2-replica K3s 3D printing portal & model vault (3dprinting.ertechlab.com) + 58 open-source PowerShell Intune/Autopilot proactive remediation modules.

## // 05 INDUSTRY CERTIFICATIONS & TECHNICAL CREDENTIALS

✓ Microsoft Certified: Azure Administrator Associate (AZ-104)
✓ Microsoft 365 Certified: Administrator Expert (MS-102)
✓ Microsoft Entra ID: Administration & Identity Security
✓ CompTIA Network+ Certified
✓ CompTIA Security+ Certified
✓ Inter-Tel University: Telecommunications & VOIP Systems